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MASTERY



DOCKER GULDE



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Introduction

Welcome to the Docker Mastery Guide 🙌

Whether you're a novice exploring the world of Docker or a seasoned pro seeking a quick reference, you're in the right place. This comprehensive guide is designed to address both beginners and experienced users, offering a thorough exploration of essential **Docker commands and concepts**.

Explore creating and managing **Docker images**, understand **Container management**, and master **Dockerfile** syntax, dive into **Docker Compose** complexities, and learn the latest features like **Docker Compose Watch** with clear explanations, step-by-step guidance, and practical illustrative examples.

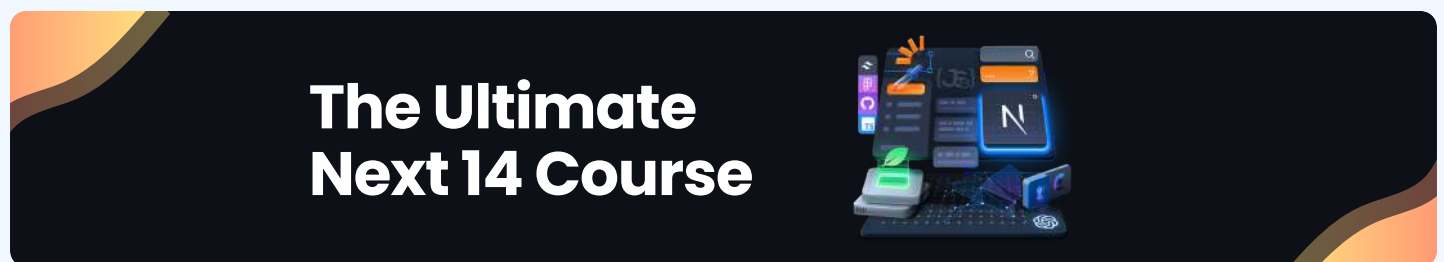
With all the necessary commands and concepts, this guide will help you get started with Docker real quickly. Go ahead and give it a read!

Happy Dockering!

...before you go

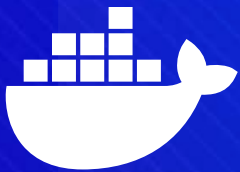
While our Docker Mastery Guide is fantastic for getting a grip on Docker and taking that first step into DevOps, imagine how much cooler it would be to apply that knowledge to the latest tech stack like Next.js and work on big projects showcasing Docker's true power.

If you're eager to dive deep into something this specific and build substantial projects, our [special course on Next.js](#) has got you covered.



It teaches everything from the ground up, providing a hands-on experience that goes beyond just Docker.

Check it out and take your skills to the next level 🚀



Docker Image

Docker Image

1. Build an image from a Dockerfile:

```
> docker build -t image_name path_to_dockerfile

# EXAMPLE

> docker build -t myapp .
```

2. List all local images:

```
> docker images

# EXAMPLE

> docker image ls
```

3. Pull an image from Docker Hub:

```
> docker pull image_name:tag

# EXAMPLE

> docker pull nginx:latest
```

Docker Image

4. Remove a local image:

```
Terminal

> docker rmi image_name:tag

# EXAMPLE

> docker rmi myapp:latest
```

Or

```
Terminal

> docker rm [image_name/image_id]

# EXAMPLE

> docker rm fd484f19954f
```

5. Tag an image:

```
Terminal

> docker tag source_image:tag new_image:tag

# EXAMPLE

> docker tag myapp:latest myapp:v1
```

Docker Image

6. Push an image to Docker Hub:

```
Terminal

> docker push image_name:tag

# EXAMPLE

> docker push myapp:v1
```

7. Inspect details of an image:

```
Terminal

> docker image inspect image_name:tag

# EXAMPLE

> docker image inspect myapp:v1
```

8. Save an image to a tar archive:

```
Terminal

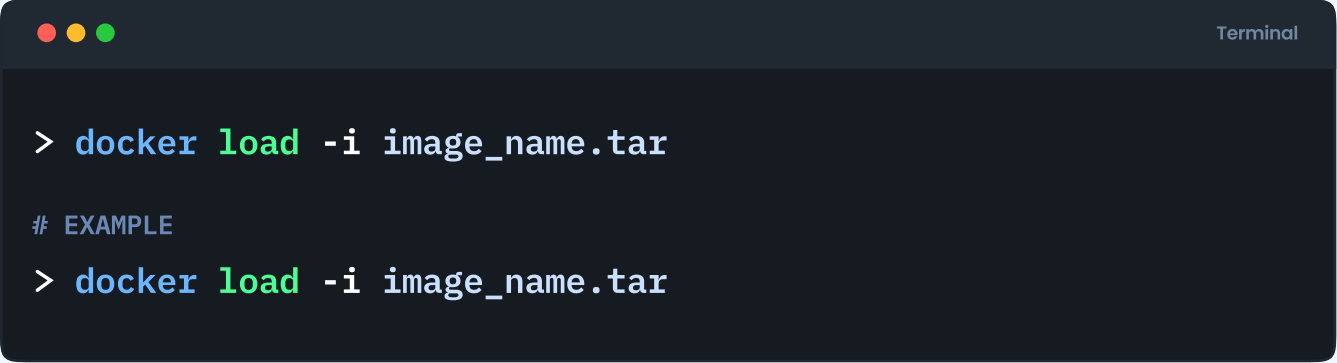
> docker save -o image_name.tar image_name:tag

# EXAMPLE

> docker save -o myapp.tar myapp:v1
```

Docker Image

9. Load an image from a tar archive:

A terminal window with a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. The word "Terminal" is in the top-right corner. The terminal contains the following text:

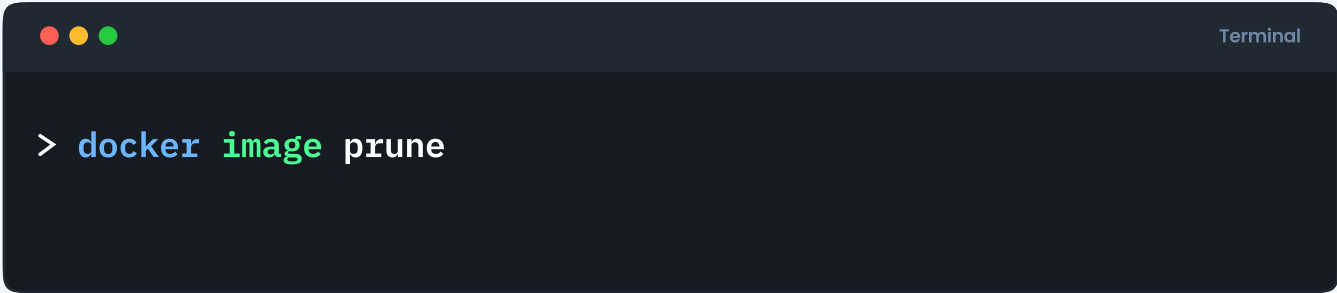
```
> docker load -i image_name.tar

# EXAMPLE

> docker load -i image_name.tar
```

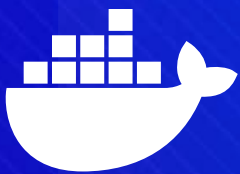
```
> docker load -i image_name.tar
```

10. Prune unused images:

A terminal window with a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. The word "Terminal" is in the top-right corner. The terminal contains the following text:

```
> docker image prune
```

```
> docker image prune
```

Docker Container

Docker Container

1. Run a container from an image:

```
Terminal

> docker run container_name image_name

# EXAMPLE

> docker run myapp
```

2. Run a named container from an image:

```
Terminal

> docker run --name container_name image_name:tag

# EXAMPLE

> docker run --name my_container myapp:v1
```

3. List all running containers:

```
Terminal

> docker ps
```

Docker Container

4. List all containers (including stopped ones):

```
> docker ps -a
```

5. Stop a running container:

```
> docker stop container_name_or_id
```

```
# EXAMPLE
```

```
> docker stop my_container
```

6. Start a stopped container:

```
> docker start container_name_or_id
```

```
# EXAMPLE
```

```
> docker start my_container
```

Docker Container

7. Run container in interactive mode:

```
Terminal

> docker run -it container_name_or_id

# EXAMPLE

> docker run -it my_container
```

8. Run container in interactive shell mode

```
Terminal

> docker run -it container_name_or_id sh

# EXAMPLE

> docker run -it my_container sh
```

9. Remove a stopped container:

```
Terminal

> docker rm container_name_or_id

# EXAMPLE

> docker rm my_container
```

Docker Container

10. Remove a running container (forcefully):

```
Terminal

> docker rm -f container_name_or_id

# EXAMPLE

> docker rm -f my_container
```

11. Inspect details of a container:

```
Terminal

> docker inspect container_name_or_id

# EXAMPLE

> docker inspect my_container
```

12. View container logs:

```
Terminal

> docker logs container_name_or_id

# EXAMPLE

> docker logs my_container
```


Docker Container

13. Pause a running container:

```
Terminal

> docker pause container_name_or_id

# EXAMPLE

> docker pause my_container
```

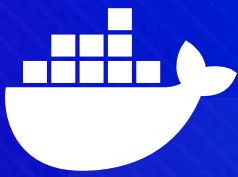
14. Unpause a paused container:

```
Terminal

> docker unpause container_name_or_id

# EXAMPLE

> docker unpause my_container
```



Docker Volumes and Network

Docker Volumes and Network

VOLUMES:

1. Create a named volume:

```
Terminal

> docker volume create volume_name

# EXAMPLE

> docker volume create my_volume
```

2. List all volumes:

```
Terminal

> docker volume ls
```

Docker Volumes and Network

3. Inspect details of a volume:

```
Terminal

> docker volume inspect volume_name

# EXAMPLE

> docker volume inspect my_volume
```

4. Remove a volume:

```
Terminal

> docker volume rm volume_name

# EXAMPLE

> docker volume rm my_volume
```

Docker Volumes and Network

5. Run a container with a volume (mount):

```
> docker run --name container_name -v volume_name:/path/in/  
container image_name:tag
```

EXAMPLE

```
> docker run --name my_container -v my_volume:/app/data myapp:v1
```

6. Copy files between a container and a volume:

```
> docker cp local_file_or_directory container_name:/path/in/  
container
```

EXAMPLE

```
> docker cp data.txt my_container:/app/data
```

Docker Volumes and Network

NETWORK (PORT MAPPING):

1. Run a container with port mapping:

```
Terminal

> docker run --name container_name -p host_port:container_port
  image_name

# EXAMPLE

> docker run --name my_container -p 8080:80 myapp
```

2. List all networks:

```
Terminal

> docker network ls
```


Docker Volumes and Network

3. Inspect details of a network:

```
> docker network inspect network_name
```

```
# EXAMPLE
```

```
> docker network inspect bridge
```

4. Create a user-defined bridge network:

```
> docker network create network_name
```

```
# EXAMPLE
```

```
> docker network create my_network
```

5. Connect a container to a network:

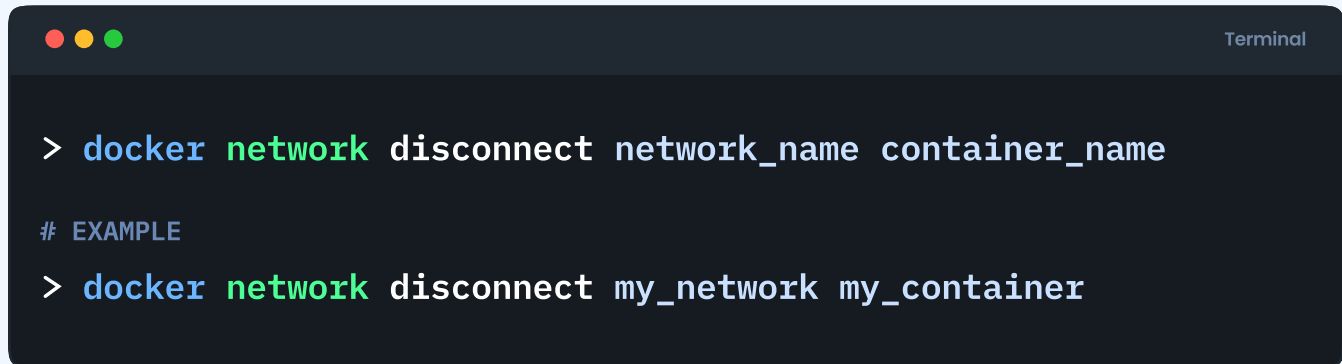
```
> docker network connect network_name container_name
```

```
# EXAMPLE
```

```
> docker network connect my_network my_container
```

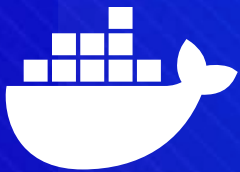
Docker Volumes and Network

6. Disconnect a container from a network:

A terminal window with a dark background and light text. The title bar shows three colored dots (red, yellow, green) on the left and the word "Terminal" on the right. The terminal contains two lines of text: a command to disconnect a container from a network, followed by an example of the same command with specific names.

```
> docker network disconnect network_name container_name

# EXAMPLE
> docker network disconnect my_network my_container
```



Docker Compose

Docker Compose

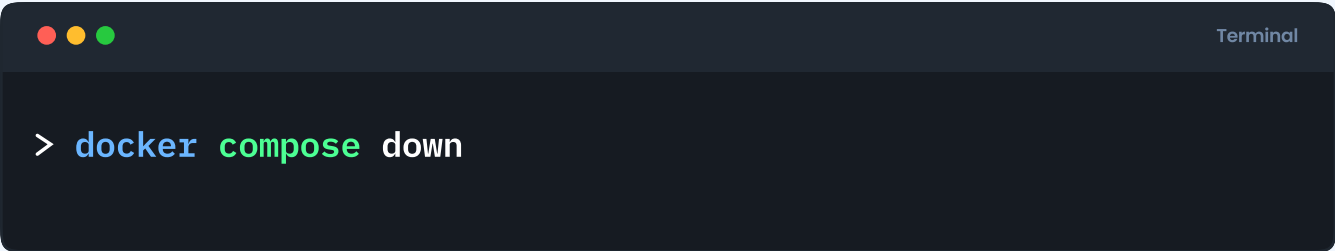
1. Create and start containers defined in a docker-compose.yml file:

A terminal window with a dark background and a title bar with three colored dots (red, yellow, green) on the left and the word "Terminal" on the right. The command `> docker compose up` is entered in the terminal.

```
> docker compose up
```

This command reads the docker-compose.yml file and starts the defined services in the background.

2. Stop and remove containers defined in a docker-compose.yml file:

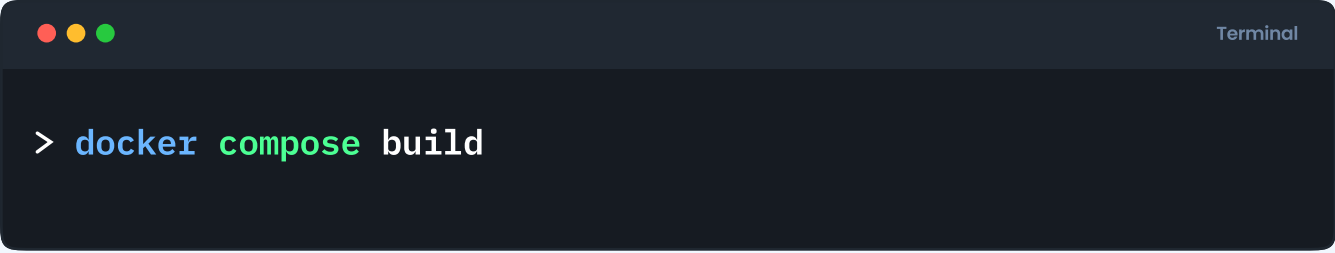
A terminal window with a dark background and a title bar with three colored dots (red, yellow, green) on the left and the word "Terminal" on the right. The command `> docker compose down` is entered in the terminal.

```
> docker compose down
```

This command stops & removes the containers, networks, and volumes defined in the docker-compose.yml file.

Docker Compose

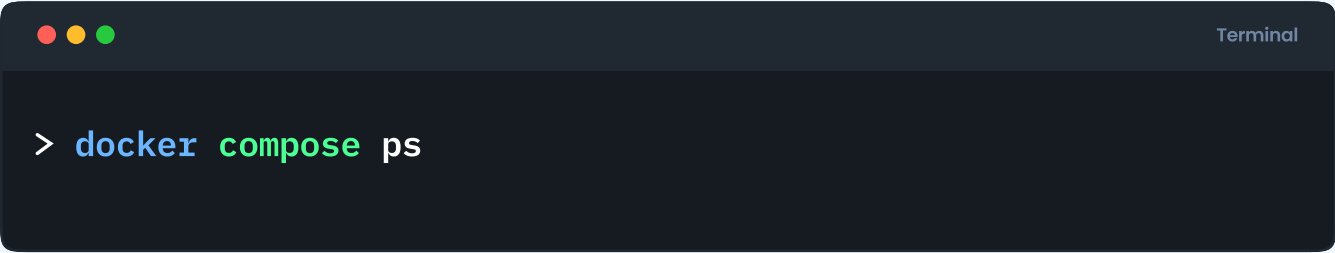
3. Build or rebuild services:

A terminal window with a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. The word "Terminal" is in the top-right corner. The command `> docker compose build` is entered in the terminal.

```
> docker compose build
```

This command builds or rebuilds the Docker images for the services defined in the `docker compose.yml` file.

4. List containers for a specific Docker Compose project:

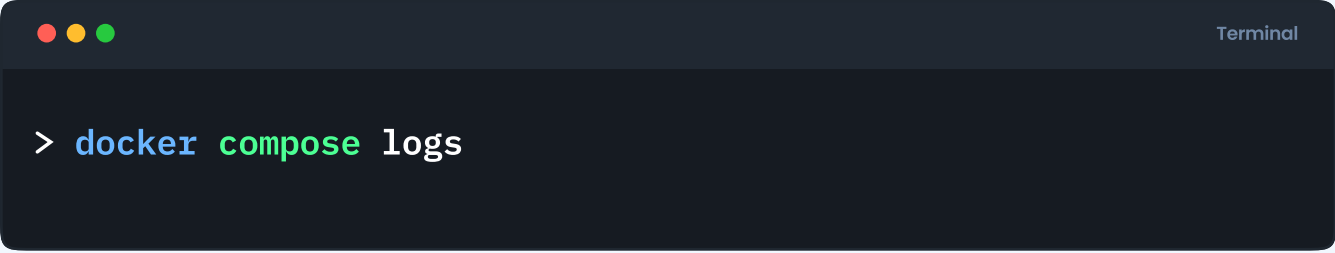
A terminal window with a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. The word "Terminal" is in the top-right corner. The command `> docker compose ps` is entered in the terminal.

```
> docker compose ps
```

This command lists the containers for the services defined in the `docker-compose.yml` file.

Docker Compose

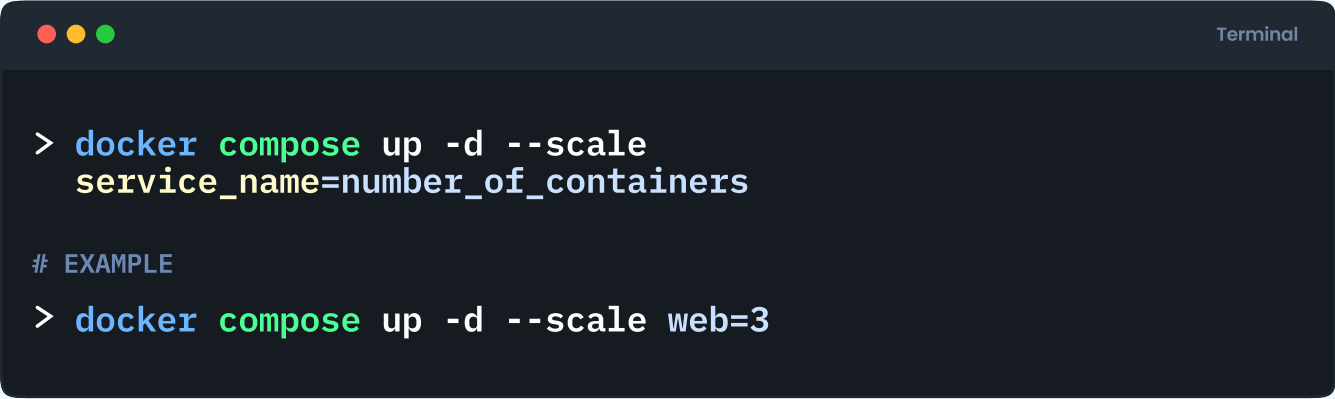
5. View logs for services:



```
> docker compose logs
```

This command shows the logs for all services defined in the docker-compose.yml file.

6. Scale services to a specific number of containers:



```
> docker compose up -d --scale  
  service_name=number_of_containers  
  
# EXAMPLE  
> docker compose up -d --scale web=3
```

Docker Compose

7. Run a one-time command in a service:

```
Terminal
> docker compose run service_name command

# EXAMPLE
> docker compose run web npm install
```

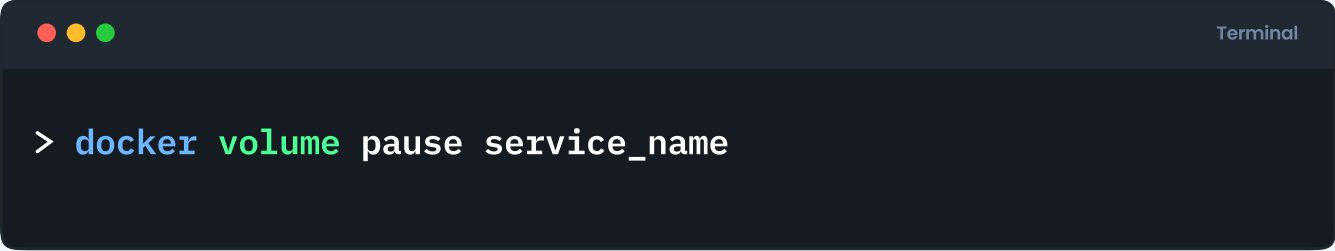
8. List all volumes:

```
Terminal
> docker volume ls
```

Docker Compose creates volumes for services. This command helps you see them.

Docker Compose

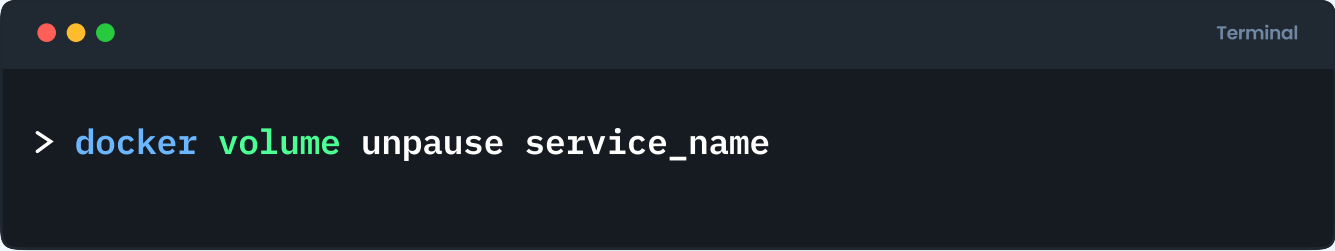
9. Pause a service:

A terminal window with a dark background and a title bar with three colored dots (red, yellow, green) on the left and the word "Terminal" on the right. The command `> docker volume pause service_name` is entered in the terminal.

```
> docker volume pause service_name
```

This command pauses the specified service.

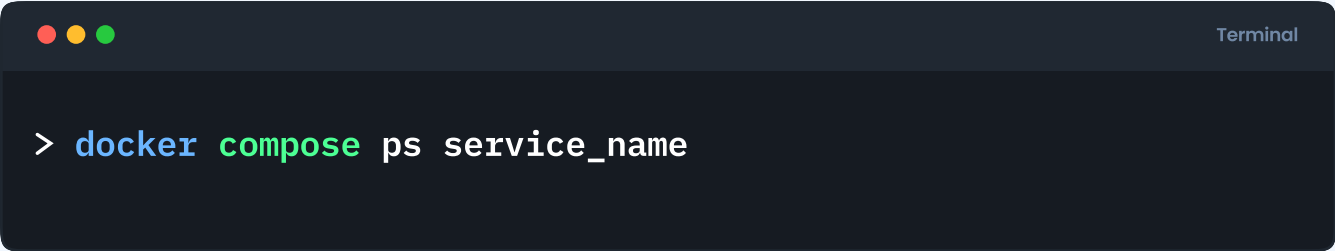
10. Unpause a service:

A terminal window with a dark background and a title bar with three colored dots (red, yellow, green) on the left and the word "Terminal" on the right. The command `> docker volume unpause service_name` is entered in the terminal.

```
> docker volume unpause service_name
```

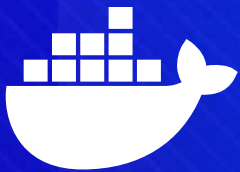
This command unpauses the specified service.

11. View details of a service:

A terminal window with a dark background and a title bar with three colored dots (red, yellow, green) on the left and the word "Terminal" on the right. The command `> docker compose ps service_name` is entered in the terminal.

```
> docker compose ps service_name
```

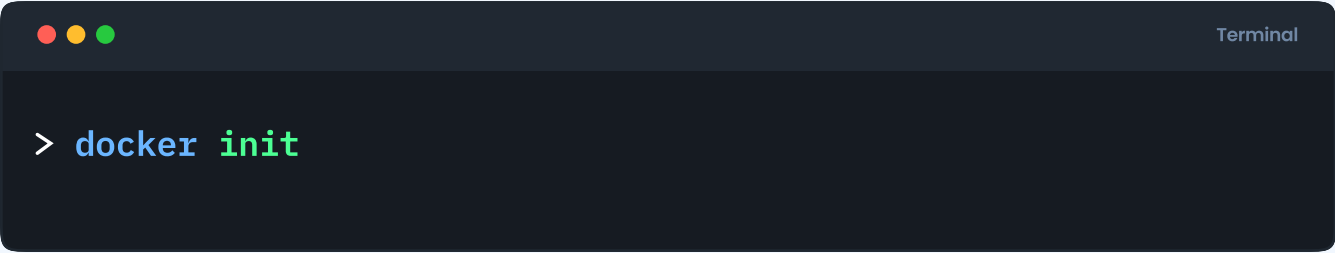
Provides detailed information about a specific service.



Latest Docker

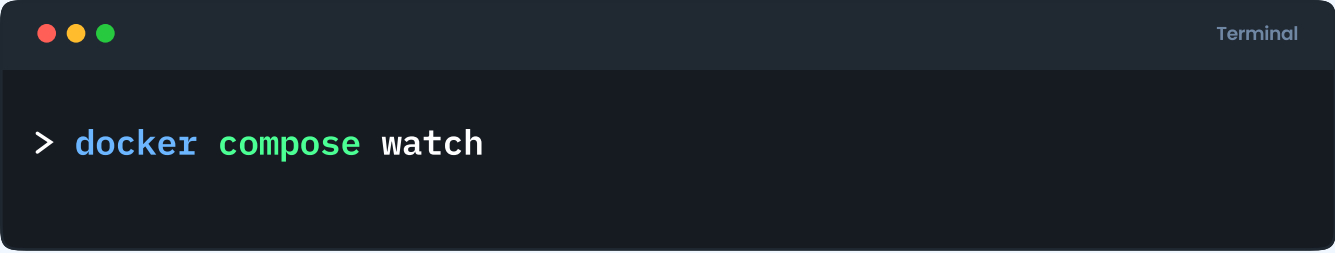
Latest Docker

1. Initialize Docker inside an application

A terminal window with a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. The word "Terminal" is in the top-right corner. The command `> docker init` is entered, with `docker` in blue and `init` in green.

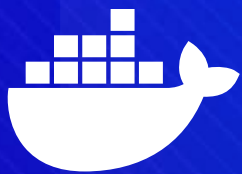
```
> docker init
```

2. Watch the service/container of an application

A terminal window with a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. The word "Terminal" is in the top-right corner. The command `> docker compose watch` is entered, with `docker` in blue, `compose` in green, and `watch` in white.

```
> docker compose watch
```

It watches build context for service and rebuild/refresh containers when files are updated



Dockerfile Reference

Dockerfile Reference

What is a Dockerfile?

A Dockerfile is a script that contains instructions for building a Docker image. It defines the base image, sets up environment variables, installs software, and configures the container for a specific application or service.

DOCKERFILE SYNTAX

FROM:

Specifies the base image for the Docker image.

```
FROM image_name:tag
```

```
# EXAMPLE
```

```
FROM ubuntu:20.04
```

Dockerfile Reference

WORKDIR:

Sets the working directory for subsequent instructions.

```
WORKDIR /path/to/directory
```

```
# EXAMPLE
```

```
WORKDIR /app
```

COPY:

Copies files or directories from the build context to the container.

```
COPY host_source_path container_destination_path
```

```
# EXAMPLE
```

```
COPY . .
```

Dockerfile Reference

RUN:

Executes commands in the shell.

```
RUN command1 && command2
```

```
# EXAMPLE
```

```
RUN apt-get update && apt-get install -y curl
```

ENV:

Sets environment variables in the image.

```
ENV KEY=VALUE
```

```
# EXAMPLE
```

```
ENV NODE_VERSION=14
```

Dockerfile Reference

EXPOSE:

Informs Docker that the container listens on specified network ports at runtime.

```
EXPOSE port
```

```
# EXAMPLE
```

```
EXPOSE 8080
```

CMD:

Provides default commands or parameters for an executing container.

```
CMD ["executable","param1","param2"]
```

```
# EXAMPLE
```

```
CMD ["npm", "start"]
```

Or,

Dockerfile Reference

```
CMD executable param1 param2
```

```
# EXAMPLE
```

```
CMD npm run dev
```

ENTRYPOINT:

Configures a container that will run as an executable.

```
ENTRYPOINT ["executable","param1","param2"]
```

```
# EXAMPLE
```

```
ENTRYPOINT ["node", "app.js"]
```

Or,

```
ENTRYPOINT executable param1 param2
```

```
# EXAMPLE
```

```
ENTRYPOINT node app.js
```

Dockerfile Reference

ARG:

Defines variables that users can pass at build-time to the builder with the docker build command.

```
ARG VARIABLE_NAME=default_value

# EXAMPLE
ARG VERSION=latest
```

VOLUME:

Creates a mount point for external volumes or other containers.

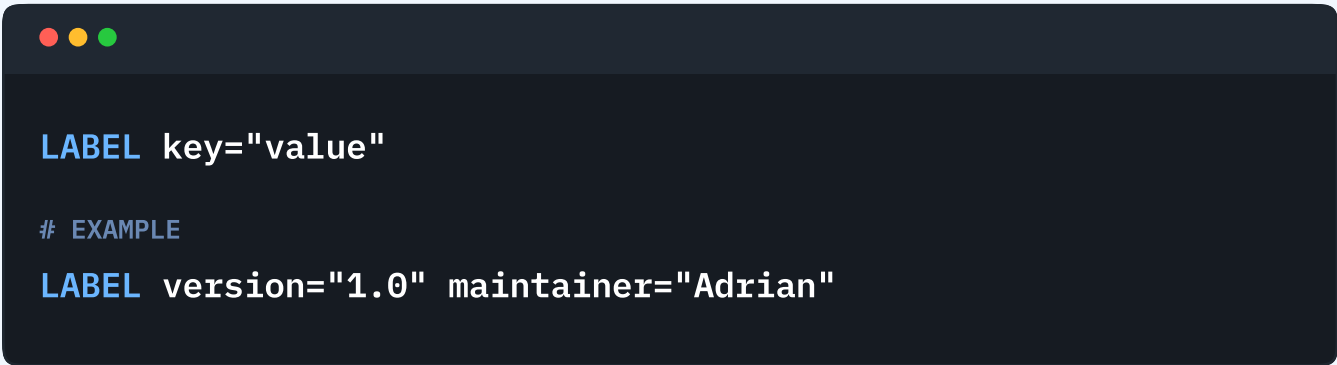
```
VOLUME /path/to/volume

# EXAMPLE
VOLUME /data
```

Dockerfile Reference

LABEL:

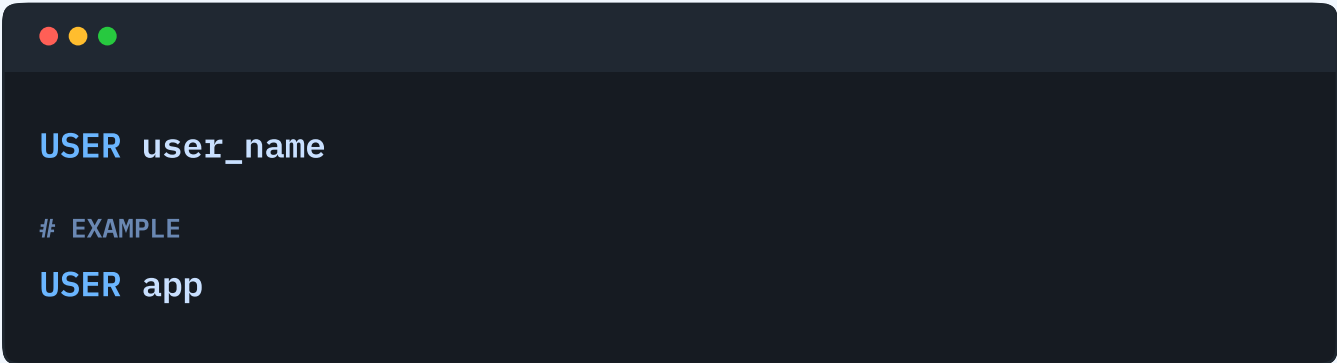
Adds metadata to an image in the form of key-value pairs.



```
LABEL key="value"  
  
# EXAMPLE  
LABEL version="1.0" maintainer="Adrian"
```

USER:

Specifies the username or UID to use when running the image.



```
USER user_name  
  
# EXAMPLE  
USER app
```

Dockerfile Reference

ADD:

Copies files or directories and can extract tarballs in the process.

```
ADD source_path destination_path
```

```
# EXAMPLE
```

```
ADD ./app.tar.gz /app
```

Similar to **COPY**, but with additional capabilities (e.g., extracting archives).

Dockerfile Example

```
# Use an official Node.js runtime as a base image
FROM node:20-alpine

# Set the working directory to /app
WORKDIR /app

# Copy package.json and package-lock.json to the
working directory
COPY package*.json ./

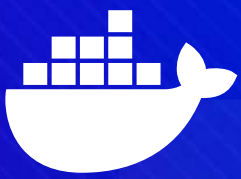
# Install dependencies
RUN npm install

# Copy the current directory contents to the container
at /app
COPY . .

# Expose port 8080 to the outside world
EXPOSE 8080

# Define environment variable
ENV NODE_ENV=production

# Run app.js when the container launches
CMD node app.js
```



Docker Compose File Reference

Docker Compose File Reference

What is a Docker Compose File?

A Docker Compose file is a YAML file that defines a multi-container Docker application. It specifies the services, networks, and volumes for the application, along with any additional configuration options.

DOCKER COMPOSE FILE SYNTAX

version:

Specifies the version of the Docker Compose file format.

Example:

```
version: '3.8'
```

Docker Compose File Reference

services:

Defines the services/containers that make up the application.

Example:

```
services:
  web:
    image: nginx:latest
```

networks:

Configures custom networks for the application.

Example:

```
networks:
  my_network:
    driver: bridge
```


Docker Compose File Reference

volumes:

Defines named volumes that the services can use.

Example:

```
volumes:  
  my_volume:
```

environment:

Sets environment variables for a service.

Example:

```
environment:  
  - NODE_ENV=production
```

Docker Compose File Reference

ports:

Maps host ports to container ports.

Example:

```
ports:  
  - "8080:80"
```

depends_on:

Specifies dependencies between services, ensuring one service starts before another.

Example:

```
depends_on:  
  - db
```

Docker Compose File Reference

build:

Configures the build context and Dockerfile for a service.

Example:

```
build:  
  context: .  
  dockerfile: Dockerfile.dev
```

volumes_from:

Mounts volumes from another service or container.

Example:

```
volumes_from:  
  - service_name
```

Docker Compose File Reference

command:

Overrides the default command specified in the Docker image.

Example:

```
command: ["npm", "start"]
```

Docker Compose File Example

Here's a simple Docker Compose file example for a web and database service:

```
version: '3.8'

# Define services for the MERN stack
services:

  # MongoDB service
  mongo:
    image: mongo:latest
    ports:
      - "27017:27017"
    volumes:
      - mongo_data:/data/db
    environment:
      MONGO_INITDB_ROOT_USERNAME: admin
      MONGO_INITDB_ROOT_PASSWORD: admin

  # Node.js (Express) API service
  api:
    build:
      # Specify the build context for the API service
      context: ./api

  # Specify the Dockerfile for building the API service
  dockerfile: Dockerfile
```

Docker Compose File Example

```
ports:
  - "5000:5000"

# Ensure the MongoDB service is running before starting
the API
depends_on:
  - mongo

environment:
  MONGO_URI: mongodb://admin:admin@mongo:27017/
mydatabase

networks:
  - mern_network

# React client service
client:
  build:
    # Specify the build context for the client service
    context: ./client
    # Specify the Dockerfile for building the client service

    dockerfile: Dockerfile

ports:
  - "3000:3000"
```

Docker Compose File Example

```
# Ensure the API service is running before starting the
client
  depends_on:
    - api

  networks:
    - mern_network

# Define named volumes for persistent data
volumes:
  mongo_data:

# Define a custom network for communication between
services
networks:
  mern_network:
```

The End

Congratulations on reaching the end of our guide! But hey, learning doesn't have to stop here.

If you're craving a more personalized learning experience with the guidance of expert mentors, we have something for you — [Our Masterclass](#).

JSM Masterclass Experience

In this special program, we do not just teach concepts – offering hands-on training, workshops, one on one with senior mentors, but also help you build production-ready applications in an industry-like environment, working alongside a team and doing code reviews with mentors. It's almost a real-world experience simulation, showcasing how teams and developers collaborate.

If this sounds like something you need, then don't stop yourself from leveling up your skills from junior to senior.

Keep the learning momentum going. Cheers! 🚀